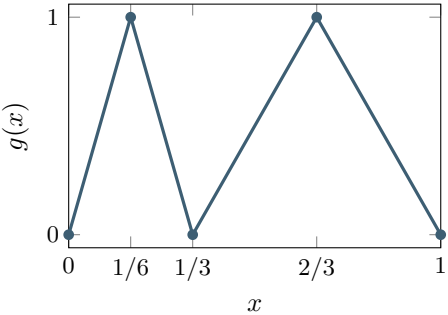
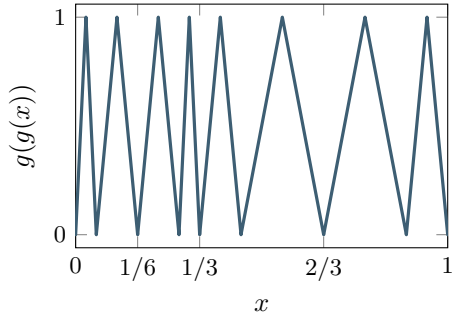
a) $g(x)$



b) $g(g(x))$